Identifying Aging Patterns in Bone Marrow Using Single-Cell RNA-seq

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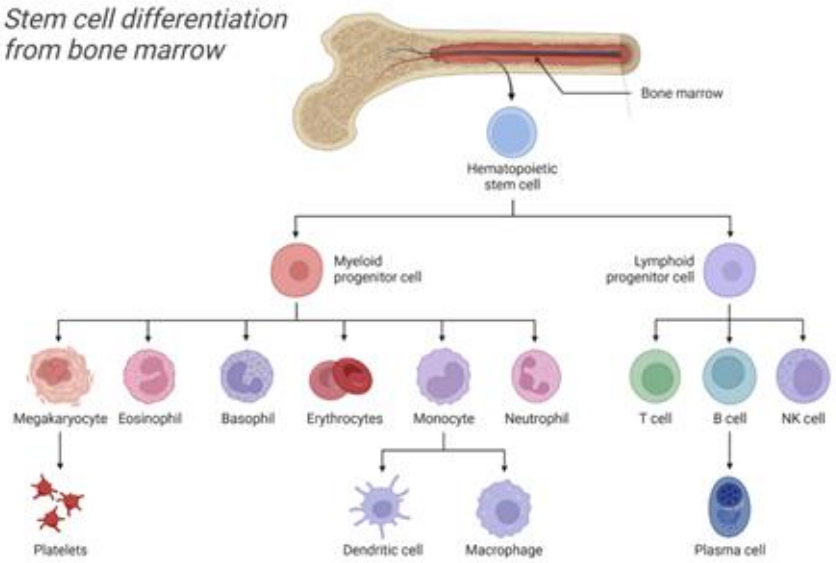
Motivation & Overview

- Aging changes bone marrow cell populations
- Bone marrow stem cells give rise to diverse immune and blood cells
- Changes are unclear at the single-cell level
- We analyze young vs old healthy donors using scRNA-seq data



Source: Image from biorender

Research Goal

Problem:

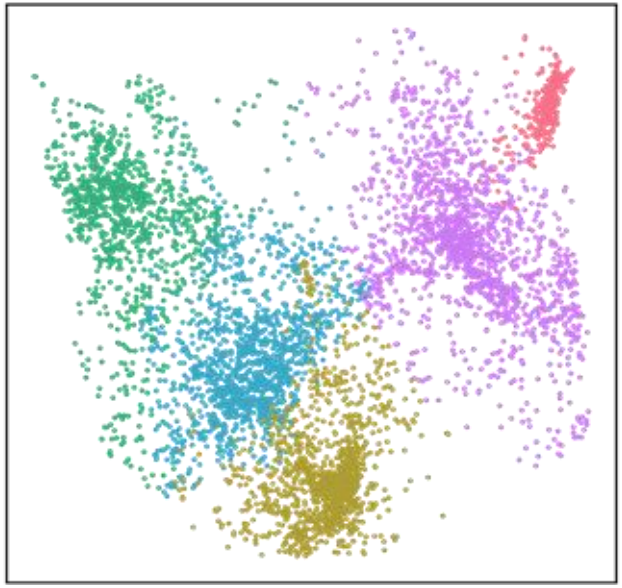
- High-dimensional single-cell data
- Difficult to identify age-related cells

Goal:

- Identify cell populations linked to aging
- Compare young vs old gene expression

Approach

- Use PCA, PHATE, Leiden and MELD
- Evaluate patterns across methods



Datasets and Related Papers

- 3 healthy young and 3 healthy old bone marrow donors profiling over 49,000 cells. This dataset provides existing cell type annotations and age group metadata, so no additional preprocessing is needed.
- This dataset was used in the paper to build single-cell proteo-genomic reference maps linking surface protein markers to cell identity across the hematopoietic system

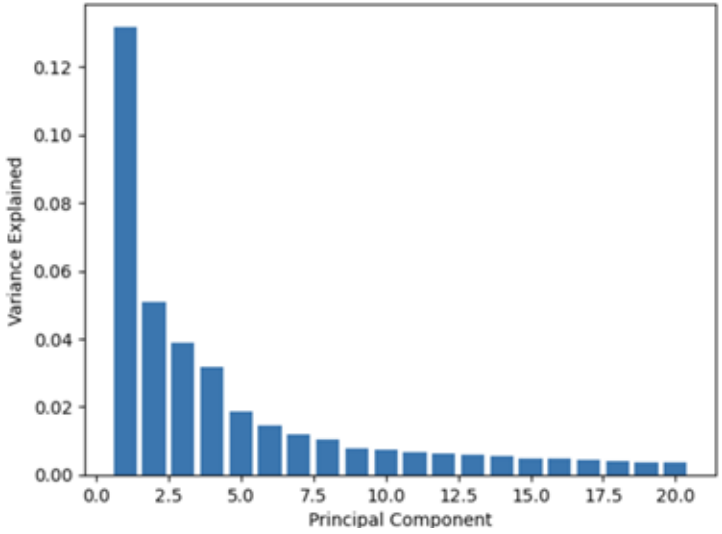
Single-cell proteo-genomic reference maps of the hematopoietic system enable the purification and massive profiling of precisely defined cell states

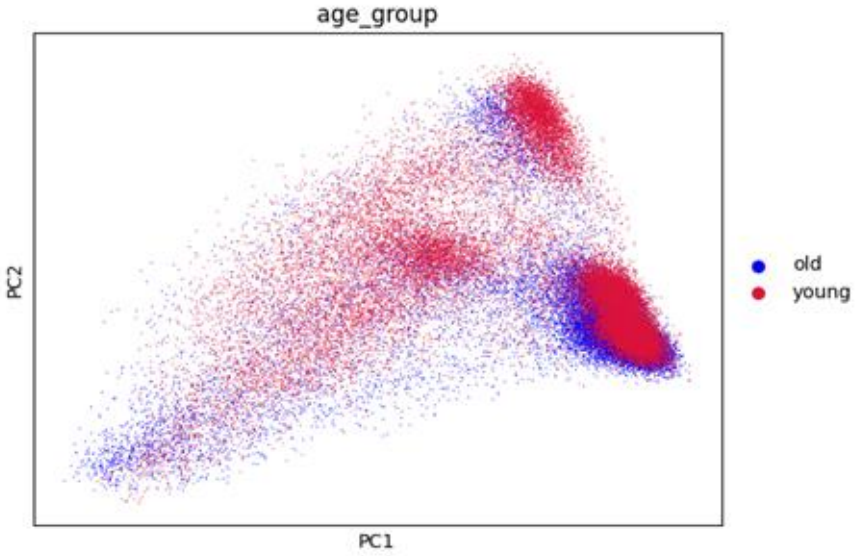
Sergio Triana, Dominik Vonficht, Lea Jopp-Saile, Simon Raffel, Raphael Lutz, Daniel Leonce, Magdalena Antes, Pablo Hernández-Malmierca, Diana Ordoñez-Rueda, Beáta Ramasz, Tobias Boch, Johann-Christoph Jann, Daniel Nowak, Wolf-Karsten Hofmann, Carsten Müller-Tidow, Daniel Hübschmann, Theodore Alexandrov, Vladimir Benes, Andreas Trumpp, Malte Paulsen, Lars Velten & Simon Haas

Nature Immunology **22**, 1577–1589 (2021) | Cite this article

69k Accesses | **183** Citations | **191** Altmetric | Metrics

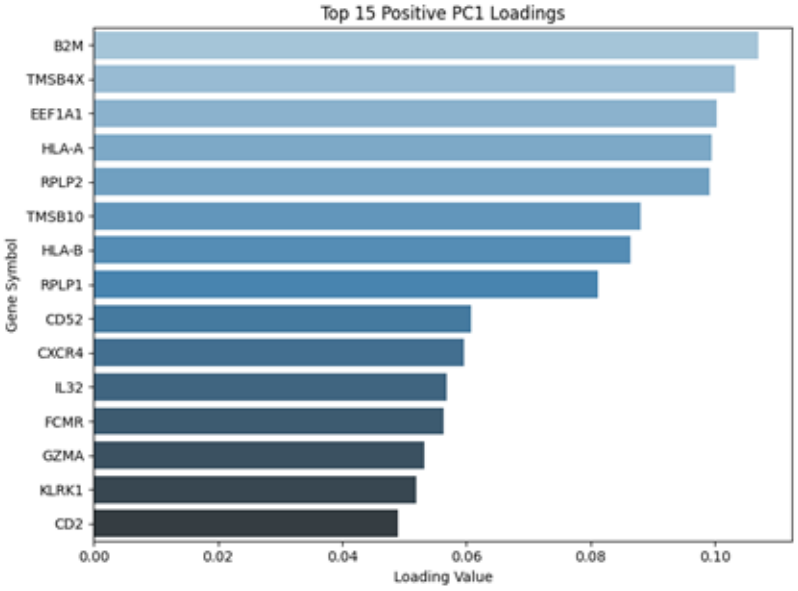
PCA shows no global separation by age

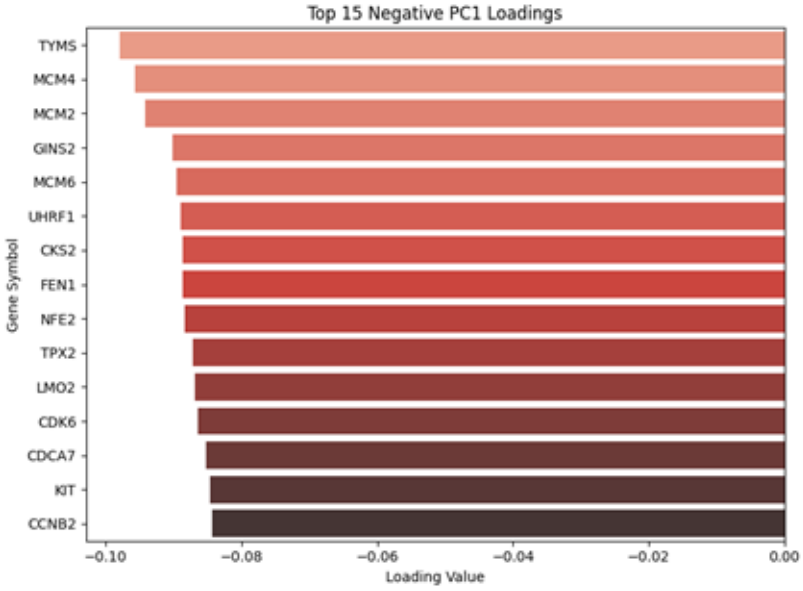




- Most variation is captured by the first few Principle components
- Young and old cells largely overlap

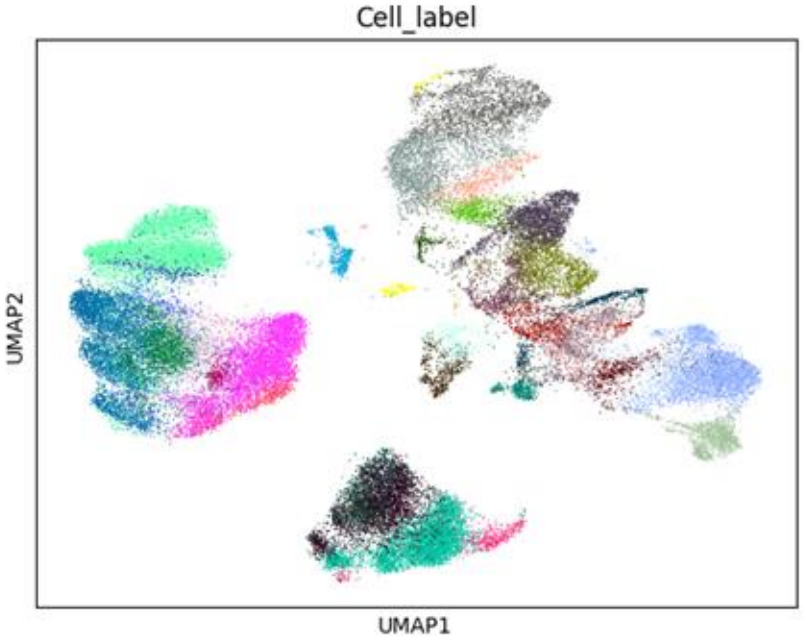
PC1 is driven by immune and cell cycle genes

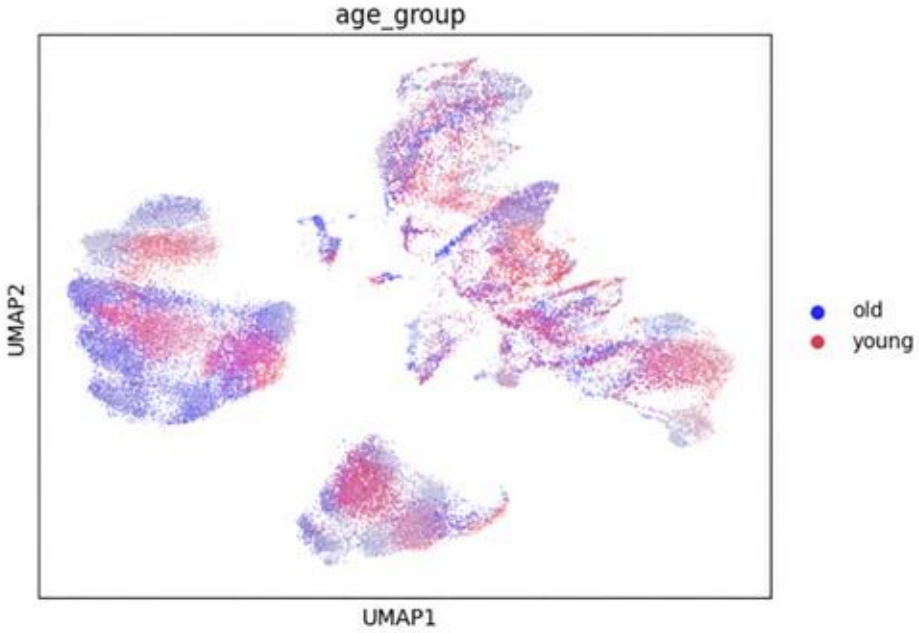




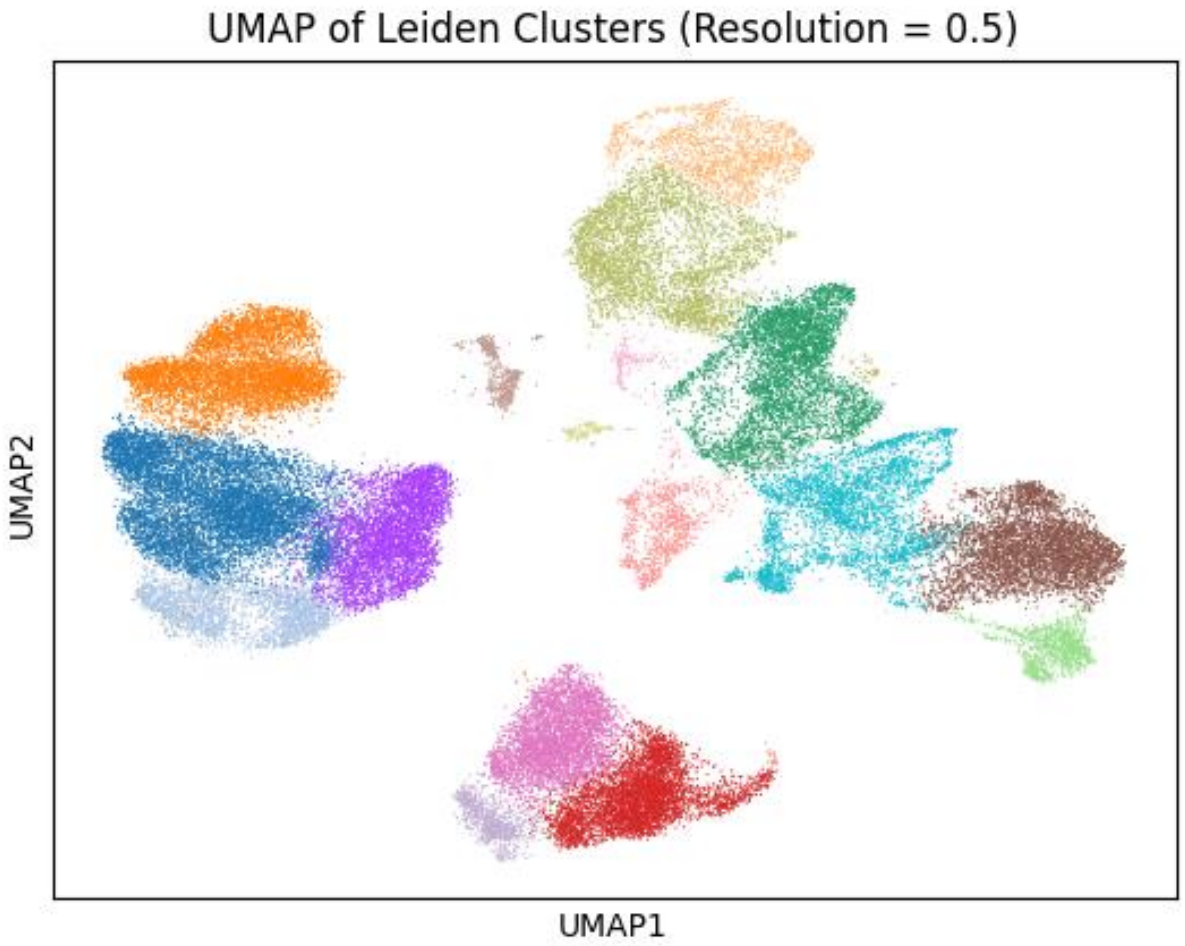
- Positive loadings (left): genes related to immune activity (eg. B2M, HLA)
- Negative loadings (right): genes related to cell growth and division (eg. MCM2, TYMS)

Cell identity drives clustering in UMAP





- Clear clustering by cell type
- Young and old cells overlap within clusters

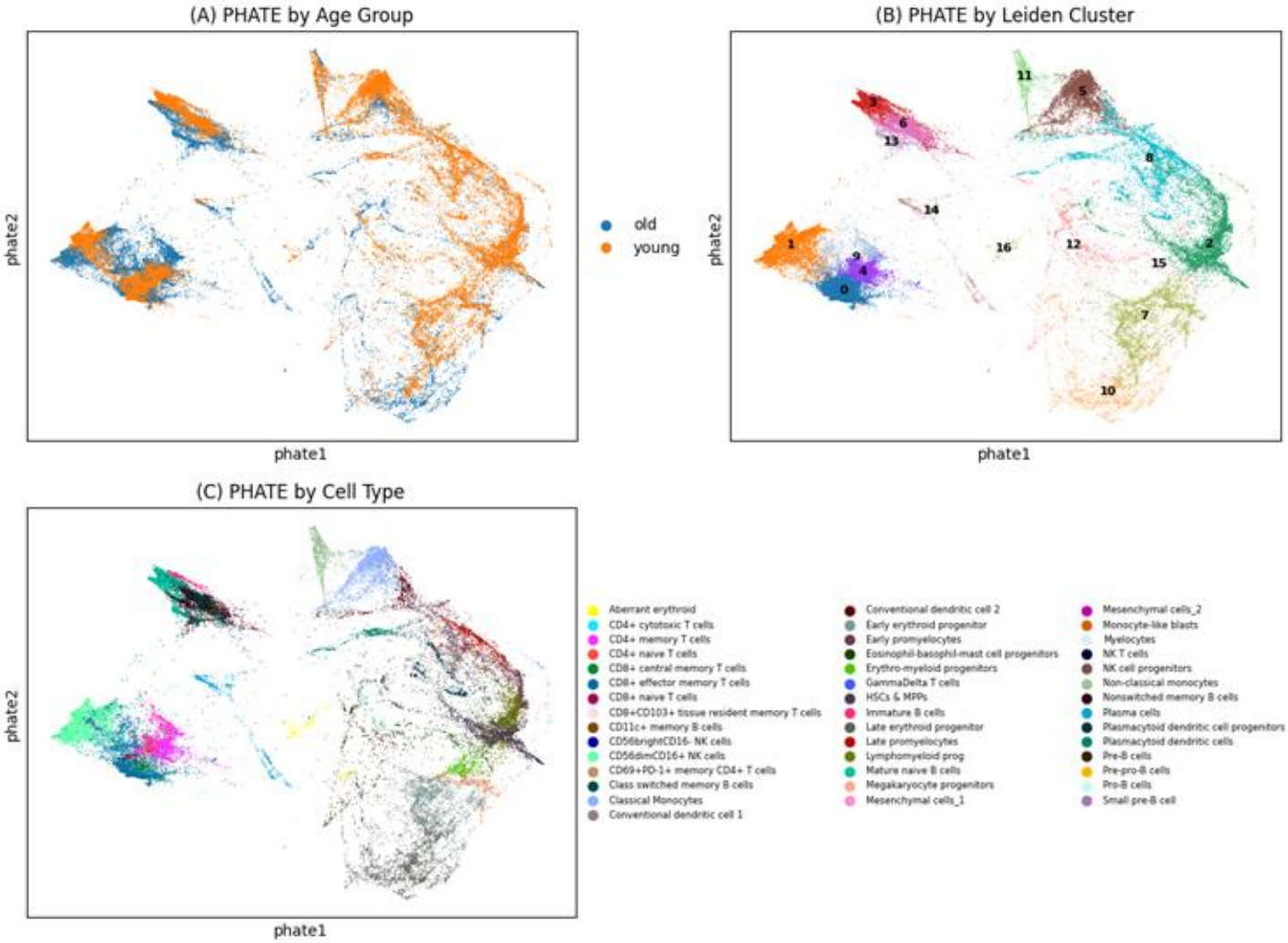


Leiden Clustering

Unsupervised Leiden clustering formed 16 clusters at resolution 0.5 on the KNN graph.

The raw clustering achieved an ARI of 0.556 and an NMI of 0.747.

- NMI suggested that the graph structure recovers most biological partitions with slight disagreements
- Underrepresented cells may have resulted in the lower ARI score.

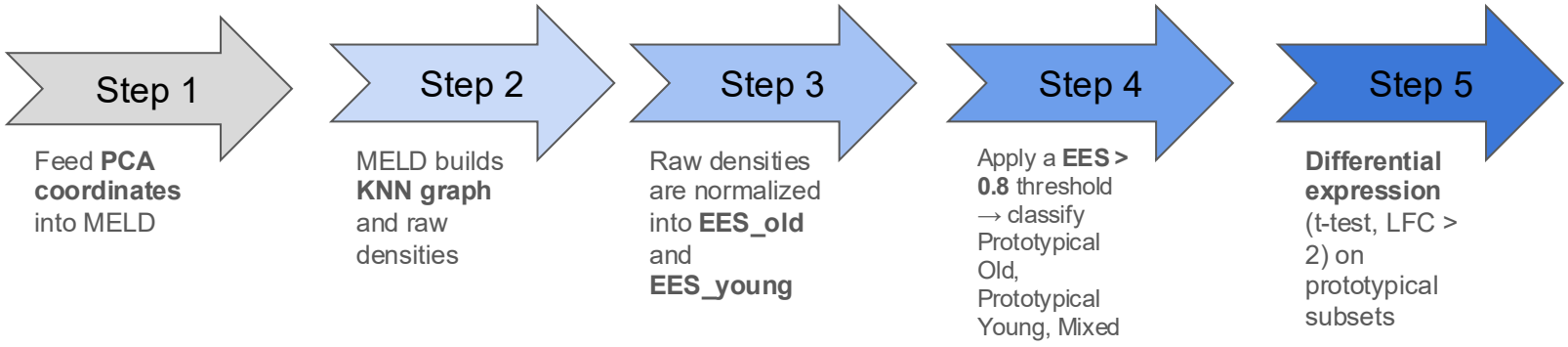


PHATE analysis

Distinct clusters appear corresponding to dominant cell populations, consistent with Leiden and cell-type coloring. However, these biologically intermediate states are made apparent by coloring PHATE by age group, as old and young cells co-exist amongst cell populations.

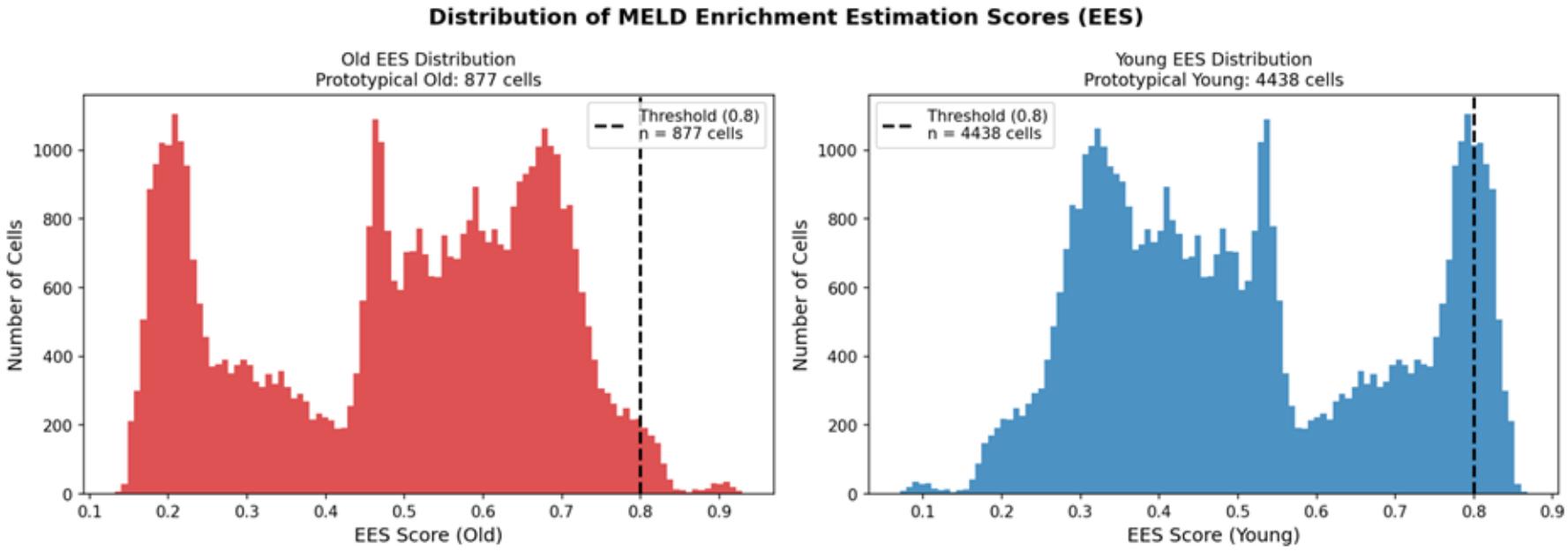
Classical monocytes and younger populations have clear subsections while most other cell types are clearly interwoven.

MELD Pipeline

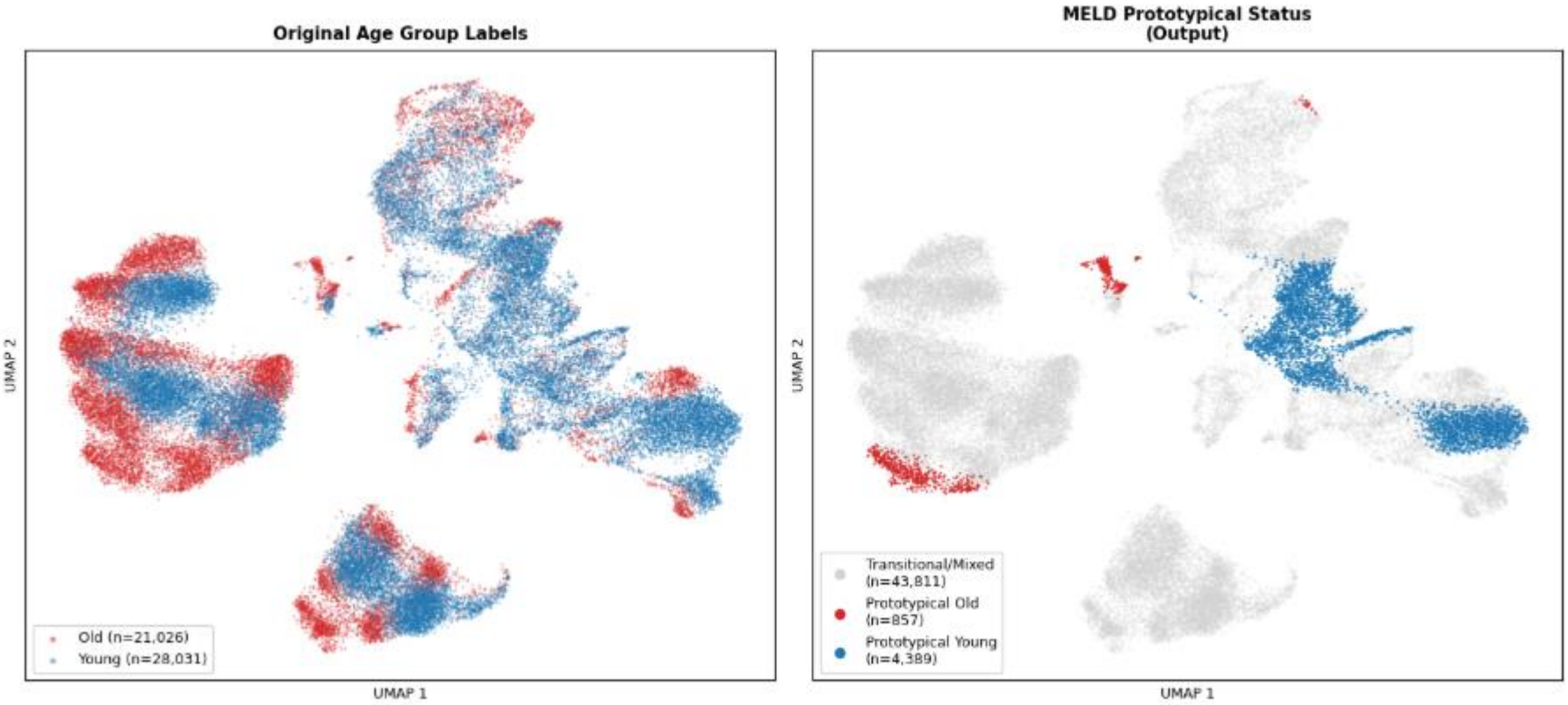


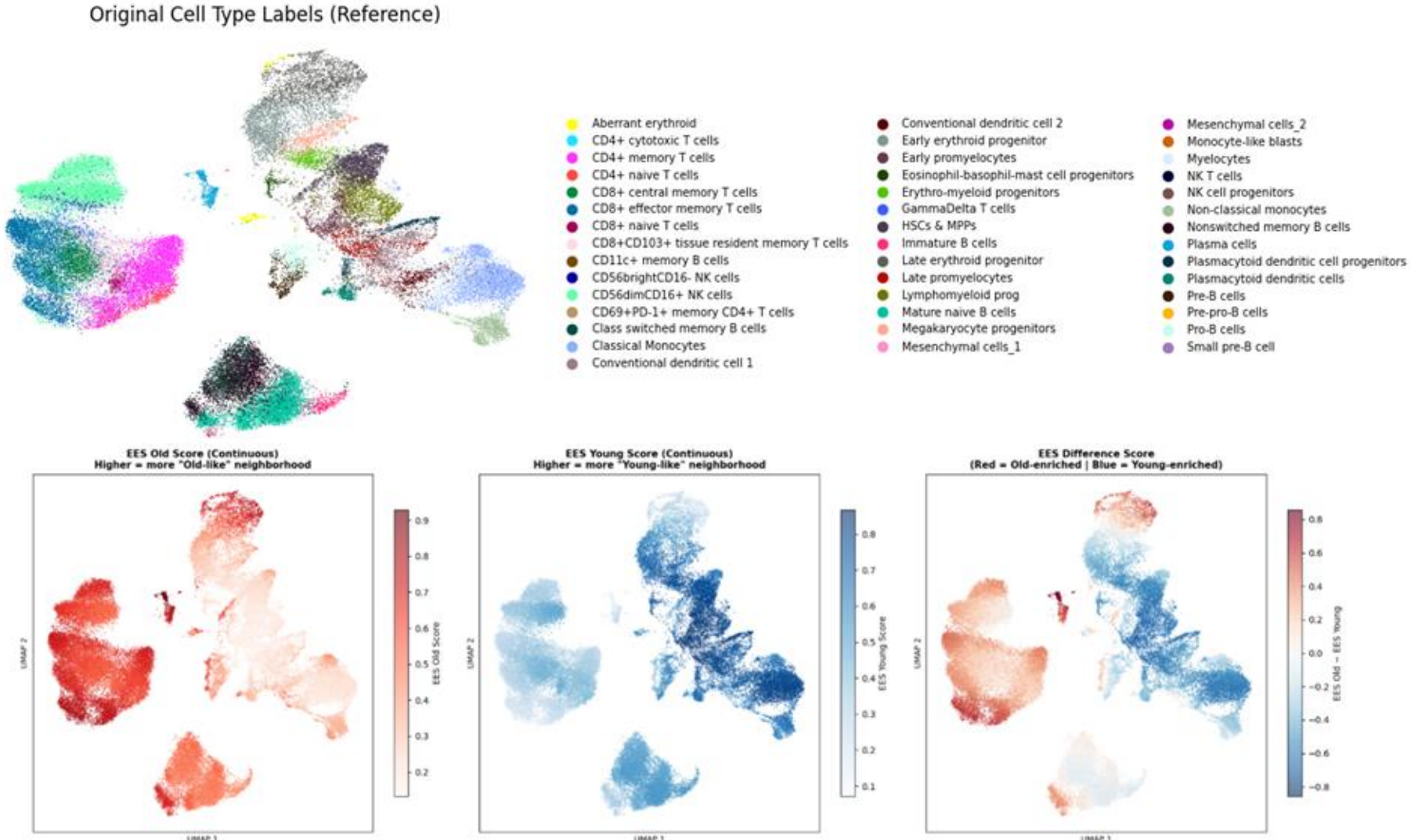
Goal: Find which cell types are most prototypical of an old bone marrow cell and a young bone marrow cell and which genes separate these two groups

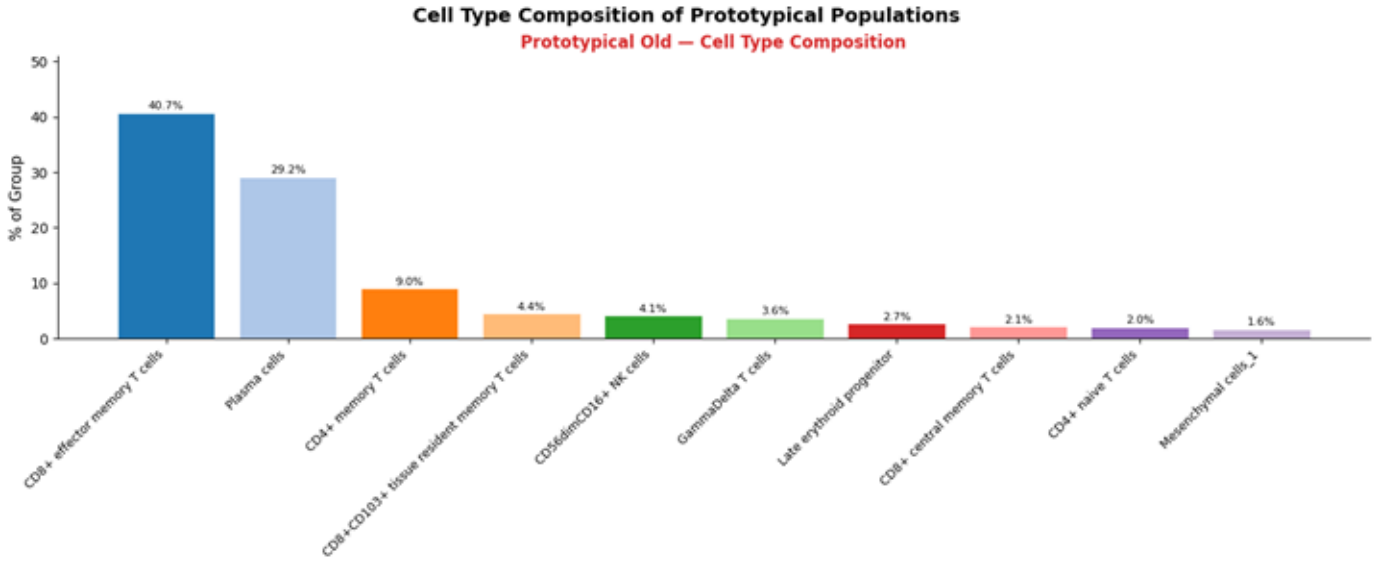
Distribution of MELD Scores

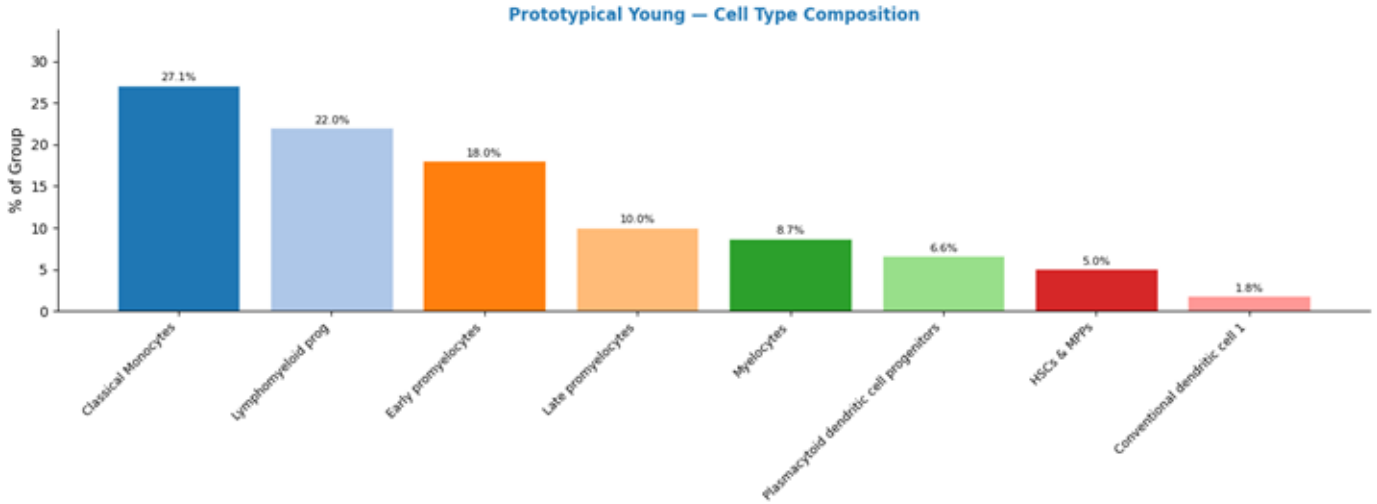


- Most Old EES scores don't surpass 0.8, resulting in a much smaller prototypical subset of cells compared to the Young EES scores
- No cell scores a Young EES score of higher than 0.9.


Original Age Group Labels
UMAP 2
UMAP 1
Old (n=21,026)
Young (n=28,031)
MELD Prototypical Status
(Output)
UMAP 2
UMAP 1
Transitional/Mixed
(n=43,811)
Prototypical Old
(n=857)
Prototypical Young
(n=4,389)


Original Cell Type Labels (Reference)
Aberrant erythroid
CD4+ cytotoxic T cells
CD4+ memory T cells
CD4+ naive T cells
CD8+ central memory T cells
CD8+ effector memory T cells
CD8+ naive T cells
CD8+CD103+ tissue resident memory T cells
CD11c+ memory B cells
CD56brightCD16- NK cells
CD56dimCD16+ NK cells
CD69+PD-1+ memory CD4+ T cells
Class switched memory B cells
Classical Monocytes
Conventional dendritic cell 1
Conventional dendritic cell 2
Early erythroid progenitor
Early promyelocytes
Eosinophil-basophil-mast cell progenitors
Erythro-myeloid progenitors
GammaDelta T cells
HSCs & MPPs
Immature B cells
Late erythroid progenitor
Late promyelocytes
Lymphomyeloid prog
Mature naive B cells
Megakaryocyte progenitors
Mesenchymal cells_1
Mesenchymal cells_2
Monocyte-like blasts
Myelocytes
NK T cells
NK cell progenitors
Non-classical monocytes
Nonswitched memory B cells
Plasma cells
Plasmacytoid dendritic cell progenitors
Plasmacytoid dendritic cells
Pre-B cells
Pre-pro-B cells
Pro-B cells
Small pre-B cell
EES Old Score (Continuous)
Higher = more "Old-like" neighborhood
EES Old Score
EES Young Score (Continuous)
Higher = more "Young-like" neighborhood
EES Young Score
EES Difference Score
(Red = Old-enriched | Blue = Young-enriched)
EES Old − EES Young
UMAP 1
UMAP 2



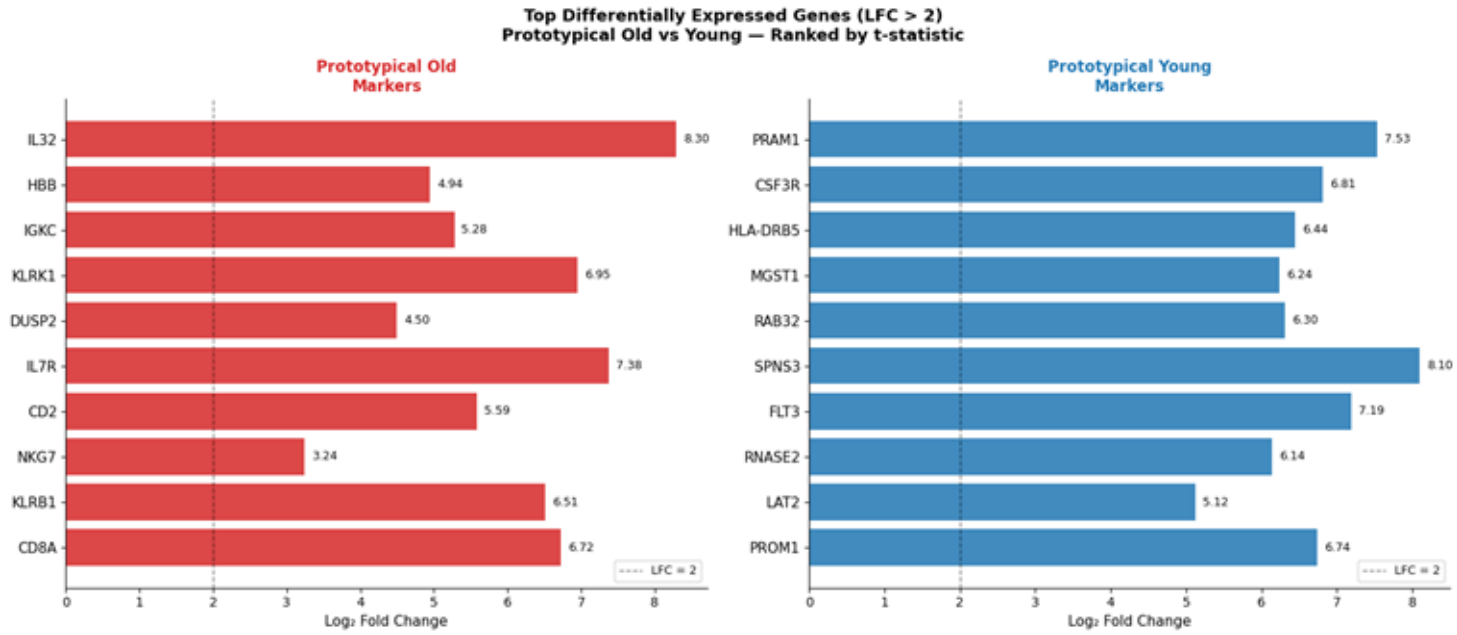


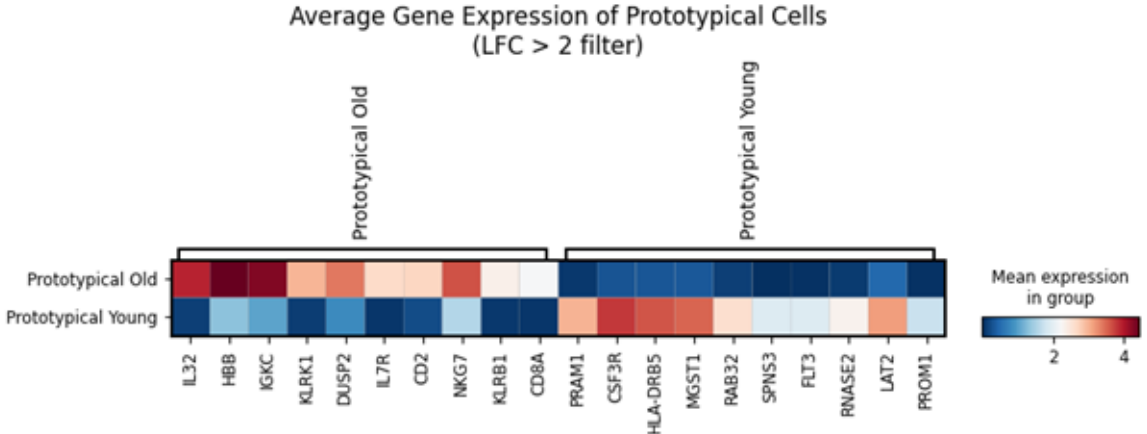
Prototypical Old (n = 877)

- Dominated by **mature lymphoid** and **immune effector** populations
- Expected with known biology that effector **memory T cells** accumulate over a lifetime

Prototypical Young (n = 4,438)

- Dominated by **myeloid** and **progenitor** populations
- Shows that **stem cells** are more productive in **young bone marrow**





Differential Expression Analysis

Prototypical Old markers included markers like **IL32** and **IL7R**, markers commonly found on T-cells. Other markers are associated with either NK cells or plasma cells.

Prototypical Young markers included markers like **PRAM1**, a gene expressed and regulated during normal myelopoiesis. Other markers are also associated with myeloid development or stem cell markers